

Software Requirements Specification

IT Service Management (ITSM) Ticketing Tool | Ref: ISO/IEC/IEEE 29148:2018

1. Introduction

1.1 Purpose

The purpose of this document is to specify the software requirements for the proposed IT Ticketing Tool. This document outlines the functional and non-functional requirements, Service Level Agreements (SLA), and operational workflows based on ITIL and ISO 20000 principles. Technical implementation details (infrastructure, programming languages, databases) have been intentionally omitted to provide flexibility for future architectural design phases.

1.2 Scope

The IT Ticketing Tool will serve as the centralized platform for logging, managing, and resolving IT-related incidents and service requests within the organization. The system is designed to streamline communication between end-users and IT support personnel, enforce SLA compliance, automate request routing, and provide comprehensive reporting capabilities.

1.3 Definitions and Acronyms

Term	Definition
SLA	Service Level Agreement; a commitment between a service provider and a client regarding measurable metrics like response and resolution times.
Incident	An unplanned interruption to an IT service or reduction in the quality of an IT service.
Service Request	A formal request from a user for something to be provided (e.g., software installation, access request).
Escalation	The process of transferring a ticket to a higher tier of support or management when SLA parameters are breached or complex intervention is required.

Standardization Notice: Requirements documented herein follow the ISO/IEC/IEEE 29148:2018 standard for requirements engineering, ensuring traceability, unambiguous definitions, and testability.

2. Overall Description

2.1 User Characteristics

The system will interact with four primary classes of users:

- **Requester (End-User):** Employees who use the system to report incidents, submit service requests, and track the status of their submitted tickets.
- **Agent (Level 1/2 Support):** IT staff responsible for receiving, analyzing, communicating with the user, and resolving standard incidents and requests.
- **Specialist (Level 3 Support):** Subject matter experts handling complex escalations that agents cannot resolve.
- **Administrator / Manager:** Overseers responsible for configuring the system, monitoring SLA compliance, generating reports, and managing user roles.

3. Service Level Agreements (SLA) Matrix

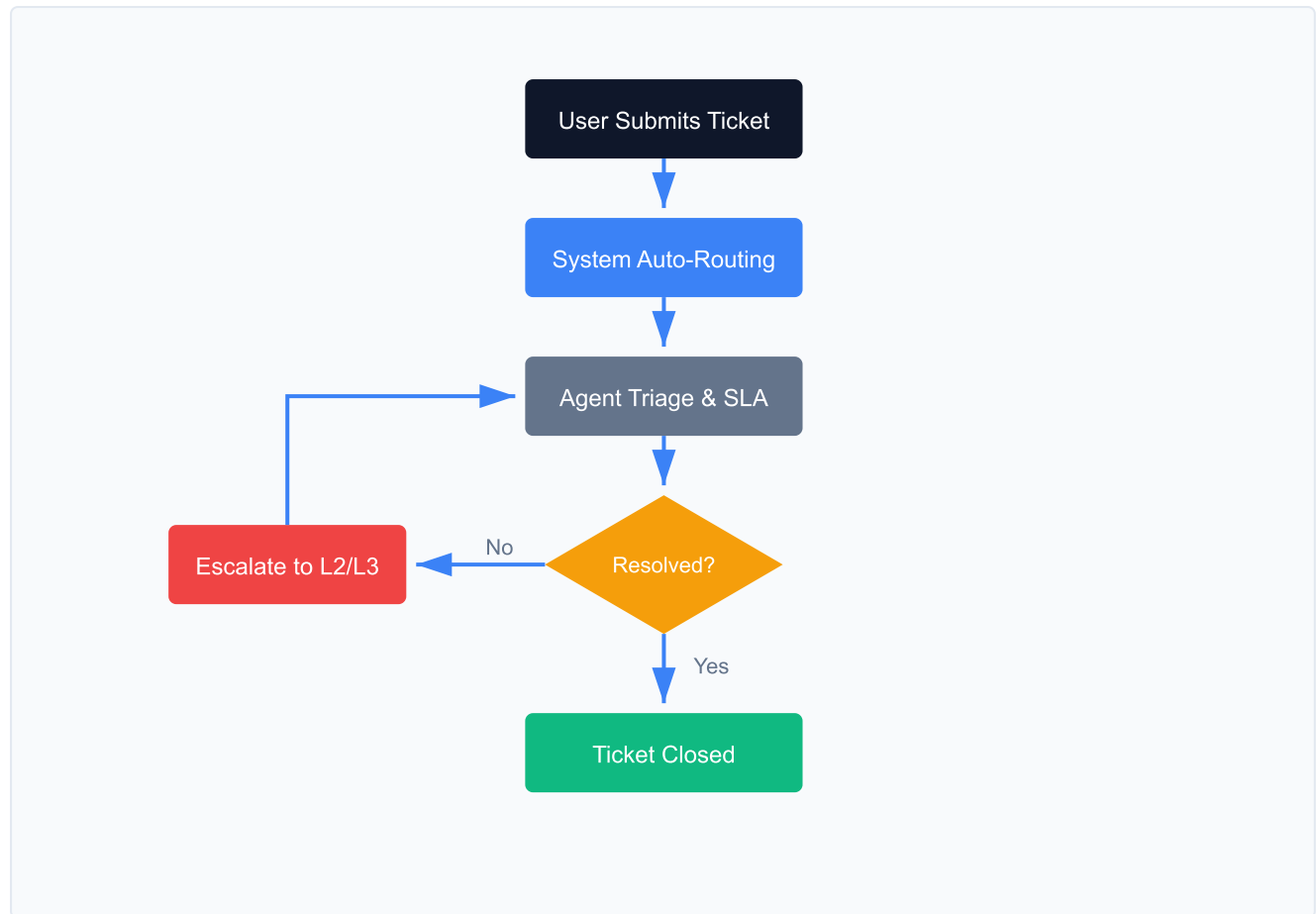
Based on ISO/IEC 20000 Service Management system requirements, the ticketing tool must automatically enforce the following SLA parameters based on predefined priority rules. The system must calculate times according to defined business hours.

Priority Level	Criteria / Impact	Target Response Time	Target Resolution Time	Escalation Trigger
P1 - Critical	Total service outage impacting a large number of users or critical business functions. No workaround available.	15 Minutes	4 Hours	Alert L3 & Management immediately; escalate if no response in 15m.
P2 - High	Significant degradation of service or major functionality loss for a group of users. Workaround is difficult.	30 Minutes	8 Hours	Escalate to L2/L3 if no resolution within 6 hours.
P3 - Medium	Single user affected, or minor feature degraded. A reasonable workaround exists.	2 Hours	24 Hours	Manager notification if nearing 20 hours unresolved.
P4 - Low	General inquiries, standard service requests, or cosmetic issues with no immediate business impact.	4 Hours	48 Hours	Automated SLA warning sent to agent at 40 hours.

Note: Response time is calculated from the moment the ticket is created until an Agent logs the first substantial interaction.

4. System Workflow Diagram

The following diagram outlines the standard operational flow of an incident or request as it passes through the ticketing platform, ensuring compliance with ITSM best practices.



4.1 Workflow Description

- 1. Creation:** The user accesses the self-service portal or sends an email to initiate a ticket. The system records the timestamp and requester details.
- 2. Auto-Routing & Categorization:** The tool categorizes the ticket based on keywords or user selection. SLA timers are immediately attached based on priority parameters.
- 3. Agent Triage:** An assigned agent reviews the ticket. If the issue is simple, it is resolved at Level 1.
- 4. Escalation (Conditional):** If the issue cannot be resolved, or SLA breach is imminent, the ticket is escalated to higher-tier support.
- 5. Resolution & Closure:** Once work is complete, the status is marked 'Resolved'. The user is notified. If the user does not contest within a designated period, the ticket is permanently 'Closed'.

5. Functional Requirements

5.1 Ticket Management and Generation

- **REQ-F01:** The system must provide a self-service portal for end-users to create, view, update, and close their own tickets.
- **REQ-F02:** The system must support omnichannel ticket creation, including email-to-ticket conversion and API-based generation from monitoring tools.
- **REQ-F03:** Every ticket must automatically generate a unique, sequential alphanumeric identifier (e.g., INC-10204).
- **REQ-F04:** The system must allow file attachments (e.g., screenshots, logs) up to a defined size limit on the ticket interface.

5.2 Routing, Assignment, and SLA Automation

- **REQ-F05:** The system must automatically assign tickets to specific agent queues based on predefined logic (e.g., category, location, or workload balancing).
- **REQ-F06:** The system must calculate and attach a specific SLA deadline to every ticket upon creation, taking into account configured business hours and holidays.
- **REQ-F07:** The SLA timer must pause when the ticket status is set to "Waiting on User" or "Pending Vendor", and resume automatically upon state change.
- **REQ-F08:** The system must support Parent-Child ticket relationships to link multiple identical incidents to a single Master Problem ticket.

5.3 Notifications and Communication

- **REQ-F09:** The system must send automated email or push notifications to users upon ticket creation, status changes, and resolution.
- **REQ-F10:** Agents must be able to add both "Public Comments" (visible to the end-user) and "Internal Notes" (visible only to IT staff).
- **REQ-F11:** The system must trigger automated alerts to management when an SLA is within 25% of breaching, and again upon an actual breach.

5.4 Reporting and Dashboards

- **REQ-F12:** The system must provide customizable dashboards displaying real-time metrics (e.g., open tickets by agent, SLA breach rate).
- **REQ-F13:** The system must be capable of generating historical reports on agent performance, ticket volume trends, and first-contact resolution rates.
- **REQ-F14:** Data exports must be available in standard formats (CSV, PDF) for external audit purposes.

6. Non-Functional Requirements

6.1 Security and Privacy (Ref. ISO/IEC 27001)

- **REQ-NF01:** The system must enforce Role-Based Access Control (RBAC), ensuring users can only view tickets they created, while agents see assigned queues.
- **REQ-NF02:** All data transmitted between the client and the application must be encrypted in transit using industry-standard protocols.
- **REQ-NF03:** The system must support Single Sign-On (SSO) integration with the organization's existing Identity Provider.

6.2 Availability and Performance

- **REQ-NF04:** The system architecture must be designed to support an uptime availability of 99.9% during standard business hours.
- **REQ-NF05:** Interface load times must not exceed 2 seconds for standard page loads and 5 seconds for complex report generation under normal concurrent load.
- **REQ-NF06:** The system must gracefully handle concurrent access by up to 500 agents and 10,000 end-users without performance degradation.

6.3 Auditing, Logging, and Usability

- **REQ-NF07:** The system must maintain an immutable, time-stamped audit trail for every action taken on a ticket, recording who made the change, what was changed, and when.
- **REQ-NF08:** The User Interface (UI) must comply with standard accessibility guidelines, ensuring it is usable by personnel with disabilities.

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